Reasoning test

For admission to Class 8

About the test

- 30 questions in 30 minutes: 10 each of maths, English and science, mixed together.
- Half the maths and science questions are from the Class 7 syllabus (the chapters listed below); the rest test reasoning. Each question is marked Basics or Class 7.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 2 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

What the test covers: basics

Basics questions test reasoning, not memory, on these topics.

Maths

1. Number patterns and series
2. Primes, factors, HCF and LCM
3. Fractions
4. Perimeter and area
5. Simple equations
6. Speed, distance and time

English

1. Vocabulary
2. Grammar
3. Sentence completion and ordering
4. Reading comprehension

Science

1. Evaporation and condensation
2. Magnets
3. Light, shadows and mirrors
4. Acids, bases and indicators
5. Physical and chemical changes
6. Heat
7. Electric circuits
8. Respiration
9. Photosynthesis
10. Motion and speed

Reasoning

1. Clocks and calendars
2. Number codes and puzzles
3. Everyday units
4. Word problems
5. Analogies, odd one out, series and codes
6. Logical deduction
7. Reading an experiment
8. Applying a principle to a new situation

Class 7 syllabus questions

Maths

1. Large Numbers Around Us
2. Arithmetic Expressions
3. A Peek Beyond the Point
4. Expressions using Letter-Numbers
5. Parallel and Intersecting Lines
6. Number Play
7. A Tale of Three Intersecting Lines

Science

1. Exploring Substances: Acidic, Basic and Neutral
2. Electricity: Circuits and their Components
3. The World of Metals and Non-metals
4. Changes around Us: Physical and Chemical

Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS · BASICS
   Find the next number in the series: 7, 10, 16, 25, 37, ___

   A 49 **B** 52
   C 55 **D** 61

2. ENGLISH · BASICS
   The old bridge was too fragile to carry the loaded truck. As used here, "fragile" most nearly means

   A ancient **B** narrow
   C weak **D** crowded

3. SCIENCE · BASICS
   Which of these will a magnet attract?

   A a copper wire **B** an iron pin
   C aluminium foil **D** a glass bead

4. MATHS · CLASS 7
   How many thousands make one lakh?

   Type-in answer: _____

5. ENGLISH · BASICS
   Choose the correct word: Each of the players ___ given a new jersey.

   A were **B** was
   C are **D** have

6. SCIENCE · CLASS 7
   Which of these turns red litmus paper blue?

   A Fresh lemon juice **B** White vinegar
   C Baking soda solution **D** Sour orange juice

7. MATHS · BASICS
   What is the largest number that divides both 48 and 72 exactly?

   A 24 **B** 36
   C 12 **D** 6

8. ENGLISH · BASICS
   Pen is to writer as brush is to ___

   A painter **B** canvas
   C colour **D** wall

9. SCIENCE · BASICS
   Which of these tastes sour because it contains an acid?

   A vinegar **B** soap solution
   C baking soda solution **D** chalk powder

10. MATHS · CLASS 7
    What is the value of 20 − 12 ÷ 4?

    A 2 **B** 17
    C 8 **D** 23

11. ENGLISH · BASICS
Choose the word most nearly opposite in meaning to "ancient".

A modern
B old
C large
D famous

12. SCIENCE · CLASS 7
Which of these elements is a non-metal?

A Iron
B Sulphur
C Copper
D Aluminium

13. MATHS · BASICS
A square garden has a perimeter of 48 m. What is its area, in square metres?

Type-in answer: _____

14. ENGLISH · BASICS
Choose the correct word: She is very fond ___ music.

A with
B of
C for
D at

15. SCIENCE · BASICS
A ball rolls 45 m across flat ground in 9 seconds at a steady speed. What is its speed, in metres per second?

Type-in answer: _____

16. MATHS · CLASS 7
Which decimal is equal to the fraction 3/4?

A 0.34
B 3.4
C 0.75
D 0.43

17. ENGLISH · BASICS
Bird is to nest as bee is to ___

A honey
B flower
C hive
D wing

18. SCIENCE · CLASS 7
Which of these changes produces a new substance?

A Tearing paper
B Dissolving salt
C Burning wood
D Melting wax

19. MATHS · BASICS
Riya cycles to school at 12 km/h and reaches in 25 minutes. How far is her school from home?

A 3 km
B 4 km
C 5 km
D 6 km

20. ENGLISH · BASICS
What comes next in the series? AZ, BY, CX, DW, ___

A DV
B EU
C FX
D EV

21. SCIENCE · BASICS
Two identical bulbs are connected in parallel to a battery and both glow. If one bulb is unscrewed, the other bulb

A goes out
B glows dimmer
C stays lit as before
D flickers and fails

22. MATHS · CLASS 7

Two angles together make a straight line. One is three times the other. What is the smaller angle, in degrees?

Type-in answer: _____

23. ENGLISH · BASICS

Although the path was steep, the cadets ___ the summit before noon.

A avoided
B lost
C feared
D reached

24. SCIENCE · CLASS 7

A circuit has a cell, a bulb and a switch. The switch is closed but the bulb does not glow. What is the most likely cause?

A The bulb's filament is broken
B The switch is in the on position
C The wires are made of copper
D The cell has just been bought

25. MATHS · BASICS

Asha spends one third of her pocket money on books and one quarter of what is left on snacks. She has ₹150 left. How much pocket money did she have?

A ₹200
B ₹250
C ₹300
D ₹360

26. ENGLISH · BASICS

Some pilots are engineers. All engineers are graduates. Which statement must be true?

A All pilots are graduates
B No pilot is a graduate
C Some pilots are graduates
D All graduates are pilots

27. SCIENCE · BASICS

A leaf on a plant is partly covered with black paper for two days, then tested with iodine. Only the uncovered part turns blue-black. This shows that

A iodine reacts with the black paper
B the covered part lost all its water
C the leaf breathes only where it is uncovered
D starch is made only where light reaches

28. MATHS · CLASS 7

If n is an odd number, which of these is always even?

A n + 2
B 3n
C n + 5
D 2n + 1

29. ENGLISH · BASICS

In a row of children, Asha is 7th from the left and 12th from the right. How many children are in the row?

A 17
B 18
C 19
D 20

30. SCIENCE · CLASS 7

Antacid tablets relieve acidity in the stomach because they

A are acidic and add acid
B kill the germs in the stomach
C are sweet and soothe pain
D are basic and neutralise acid

Answers

1	**B. 52**	The gaps grow by 3 each time: +3, +6, +9, +12, so the next gap is +15, and 37 + 15 = 52.
2	**C. weak**	Fragile means easily broken. The sentence says the bridge could not carry the weight, which points to weakness, not age or width.
3	**B. an iron pin**	Magnets attract iron and a few other metals. Copper and aluminium are metals but are not attracted, and glass is not a metal.
4	**100**	One lakh is 1,00,000 and one thousand is 1,000, so a lakh is 100 thousands. This is a type-in question.
5	**B. was**	The subject is "each", which is singular, so the verb is "was". "Players" sits nearby but is not the subject.
6	**C. Baking soda solution**	Bases turn red litmus blue, and baking soda solution is basic. Lemon juice, vinegar and orange juice are acidic.
7	**A. 24**	48 = 24 × 2 and 72 = 24 × 3, and 2 and 3 share no common factor, so 24 is the largest. 36 does not divide 48.
8	**A. painter**	A pen is the tool a writer works with; a brush is the tool a painter works with. The other options are things a brush touches.
9	**A. vinegar**	Vinegar contains acetic acid, which gives it a sour taste. Soap and baking soda are bases, and chalk is neither sour nor acidic.
10	**B. 17**	Division comes before subtraction: 12 ÷ 4 = 3, and 20 − 3 = 17. Working left to right gives (20 − 12) ÷ 4 = 2.
11	**A. modern**	Ancient means belonging to the distant past, so its opposite is modern. "Old" means nearly the same as ancient.
12	**B. Sulphur**	Sulphur is a brittle yellow solid that does not conduct electricity: a non-metal. Iron, copper and aluminium are metals.
13	**144**	A square has four equal sides, so each side is 48 ÷ 4 = 12 m, and the area is 12 × 12 = 144 square metres. This is a type-in question.
14	**B. of**	"Fond" is followed by "of": fond of music, fond of reading. The other prepositions are not used with fond.
15	**5**	Speed is distance divided by time: 45 m ÷ 9 s = 5 m/s. This is a type-in question: the answer is typed as a number.
16	**C. 0.75**	3/4 = 75/100 = 0.75. 0.34 just writes the digits 3 and 4 after the point, which is the usual mistake.
17	**C. hive**	A nest is where a bird lives; a hive is where a bee lives. Honey is what bees make and flowers are where they feed.
18	**C. Burning wood**	Burning turns wood into ash, smoke and gases, new substances that cannot become wood again. The others change only shape or state.

19	**C. 5 km**	At 12 km/h she covers 1 km every 5 minutes, so in 25 minutes she covers 5 km. The minutes have to be turned into hours first.
20	**D. EV**	The first letter moves forward one step (A, B, C, D, E) while the second moves back one step (Z, Y, X, W, V), giving EV.
21	**C. stays lit as before**	In a parallel circuit each bulb has its own path to the battery, so removing one leaves the other's path complete and unchanged.
22	**45**	Angles on a straight line add to 180°: x + 3x = 180, so x = 45° (and the other is 135°). This is a type-in question.
23	**D. reached**	"Although" signals a contrast with the difficulty: despite the steep path, they succeeded. Only "reached" completes that contrast.
24	**A. The bulb's filament is broken**	A broken filament leaves a gap in the circuit, so no current flows. A closed switch, copper wires and a new cell all help the bulb glow.
25	**C. ₹300**	After books, 2/3 is left; snacks take a quarter of that, leaving 3/4 of 2/3, which is 1/2. Half is ₹150, so she had ₹300. ₹360 comes from taking a quarter of the whole.
26	**C. Some pilots are graduates**	The pilots who are engineers must be graduates, so at least some pilots are graduates. Nothing is said about the other pilots.
27	**D. starch is made only where light reaches**	Iodine turns blue-black where there is starch. Only the part that received light made starch, so the leaf needs light to make food.
28	**C. n + 5**	Odd + odd is even, and 5 is odd, so n + 5 is always even. n + 2 and 3n stay odd, and 2n + 1 is odd for every n.
29	**B. 18**	Counting from both ends counts Asha twice, so the row has 7 + 12 − 1 = 18 children. Adding without subtracting gives 19.
30	**D. are basic and neutralise acid**	Too much stomach acid causes acidity. An antacid is a mild base, such as milk of magnesia, which neutralises the extra acid.